

Character Shader (Texture, Shading)

Basically, Toon Shader+Outline.

Effect:

 image-20240921-052042.png

Outline Effect:

Mesh change: Duplicate the base mesh and set it as the child object of the main mesh.

 image-20240921-050929.png

 image-20240921-051338.png

Outline Shader:

 image-20240921-051424.png

Toon Shader:

-Texture:

Base Color, Normal, Shadow Base Color, Shadow Intensity Map, Shine.

(Shadow Base Color stands for how much intensity each part is being influenced by shadow)

 image-20240921-054018.png  image-20240921-054842.png

>Base Color&Shine created in Substance Painter, the rest modified in Photoshop.

Experience: Set Metal Objects(Or reflective objects needed Shine) In the SAME UV TEXTURE.

Result:

 image-20240921-061349.png

>SP:

Add Normal Detail onto Diffuse map: use Fill Layer+Generator: UV Borders+Anchor Points(of the Normal)

(Turn on Micro Details and reference Anchor Point in Generator Properties

(choose Nrm/Height for channel))

 image-20240921-063822.png

>Texture Darkening for shape



>Before Exporting to Unreal, 1. delete all unused materials in Blender, 2. separate mesh that has two materials into separate material mesh



>The primary params that determined the shader artistic effect:

1. **Shader Intensity&Tint** (How dark is the shaded part, and color)
 2. **Base Shade Light Color**: Color Tendency on the lit part. (Related to Shadow Subtract above)
 - 3.**Shadow Depth**(Angle of the calculated shadow, result in proportion change of Light and Dark)
- Rest of the params for subtle adjustment

-Unreal5 Shader Referenced from

<https://www.youtube.com/watch?v=0GyaYoV6DIs>

-Customized part 1: Universal Shadow Map



>Special Materials scenarios: For example, emissive eyes effect. Create an original version with Shadow Base Texture.



>Naming Convention note: Separate different custom parent shaders.



-Troubleshooting: Shine Map doesn't give enough light in eyes



Directional Light



>Connecting DirectLight with the Toon Character:
In Character Blueprint:

